

Midhun Chakkaravarthy K

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EXPERIENCE

- Tata Elxsi** Bengaluru, India
Software Engineer June 2025 – Present
 - Developed the **content catalog & metadata service**, designing **REST APIs** for ingestion, normalization, and retrieval of content feeds; structured service to support **multi-device playback workflows and dynamic content discovery**.
 - Implemented an **ad insertion pipeline**, integrating VAST/VMAP-based ad workflows with backend services to enable **server-side and client-side ad delivery**, ensuring seamless playback and monetization across OTT streams.
 - Built a **user analytics & personalization service**, capturing **playback/session events** and enabling recommendation logic using **LLM/RAG concepts**, and exposing APIs for **real-time personalized content delivery**.
- Sayvai Software** Remote
Full Stack Developer June 2024 – May 2025
 - Developed a **Python/Flask**-based parametric pattern export system converting GarmentCode patterns into **SVG, PNG, PDF, and DXF** with high-accuracy geometry preservation.
 - Built a **3D garment visualization pipeline** using **Meshcapade API** and a **NiceGUI** interface for real-time pattern rendering and measurement-driven fit validation.
 - Refactored the pattern-processing backend into **three Flask microservices** for pattern parsing, format rendering, and export orchestration, improving modularity and deployment flexibility.
 - Engineered a **CMS platform** using **Next.js** and **FastAPI** for external clients to create and manage website pages and embed configurable **chat and voice agents**. Developed **6+ reusable modular components**, including dynamic routing, component-driven UI blocks, and backend APIs for page composition and deployment.
 - Built **administrative interfaces** that allow non-technical users to configure agent behavior, **memory settings**, and **deployment parameters**. Integrated **vector search**, **prompt templating**, and **RAG-style** retrieval pipelines to support contextual and customizable agent responses.
 - Designed a **configuration dashboard** for creating and managing **voice agents**, implementing structured **React UI flows** and **FastAPI endpoints** to manage core settings such as **STT, TTS, and LLM** configuration.
 - Developed **multi-agent management** features using **Supabase, SQLAlchemy, and Alembic**, enabling reliable configuration storage, structured schema updates, and **real-time data synchronization** across agent instances.
- Smartail Private Limited** Chennai, India
Product Engineer Intern December 2023 – May 2024
 - Redesigned the **admin portal** for **70+ schools** and **65,000+ students**, implementing class, subject, syllabus, and publication modules using **Angular** and **Spring Boot**.
 - Built features and resolved major issues across the stack using **Angular, Spring Boot, and Flask**; delivered **10+ UI components** with corresponding APIs and improved application stability.
 - Utilized **Docker** for consistent environments and **Kafka** for real-time data flows, enhancing system reliability.
- Smartail Private Limited** Chennai, India
Product Engineer Intern June 2022 – December 2022
 - Built an **image-extraction workflow** for subjective and objective questions, integrating **MathLive Keyboard** and **KaTeX** for accurate math rendering and extraction from images.
 - Developed multiple UI components and full pages using **Angular**, and integrated APIs with a **Spring Boot** backend to ensure consistent data flow and system functionality.

SKILLS SUMMARY

- Languages:** C, Java, Python, JavaScript, TypeScript, SQL.
- Frameworks & Libraries:** Angular, React, Next.js, Node.js, Express.js, FastAPI, Flask, Spring Boot, Tailwind CSS, LangChain, SQLAlchemy, Alembic.
- Databases:** MongoDB, PostgreSQL, MySQL, Supabase, Vector Databases.
- Tools:** AWS, Docker, Kubernetes, Git, Kafka, Postman, WebSockets, CI/CD.
- Core:** RAG, LLM-based Application Development, Prompt Engineering, Microservices Architecture, Distributed Systems, Event-Driven Architecture, REST API Design, System Design.

EDUCATION

- Coimbatore Institute of Technology** Coimbatore, India
Master of Science in Software Systems, Cumulative GPA: 8.43/10 July 2019 – May 2024